

## Project Initialization and Planning Phase

|               |               |
|---------------|---------------|
| Date          | 27 June 2026  |
| Team ID       |               |
| Project Name  | Rising Waters |
| Maximum Marks | 3 Marks       |

### Define Problem Statements (Customer Problem Statement Template):

Floods cause significant loss of life and property every year. Existing flood forecasting systems often fail to provide timely and accurate warnings, limiting the ability of authorities and communities to take preventive action. Therefore, there is a need for an intelligent flood prediction system that uses machine learning algorithms such as Decision Tree, Random Forest, KNN, and XGBoost to analyze weather parameters and predict flood risks accurately. The system will provide early warnings through a Flask-based web application, enabling better disaster preparedness, resource allocation, and public safety.

#### Example:



| Problem Statement (PS) | I am (Customer)                          | I'm trying to                                     | But                                                              | Because                                                                    | Which makes me feel                                   |
|------------------------|------------------------------------------|---------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------|
| PS-1                   | A resident living in a flood-prone area. | Protect my family and property from flood damage. | I do not receive sufficient advance warning before floods occur. | Accurate flood prediction systems are not easily accessible to the public. | Anxious and unprepared during heavy rainfall periods. |